On probabilistic prediction and information-theoretic operators

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From probability to predictive uncertainty

Throughout this poster, $H(X)$ denotes differential entropy as we assume the random variable to be continuous.

$$H(X) = \mathbb{E}_{p(x)}[-\log p(x)]$$

A predictive view

Self-supervised learning (SSL) generates supervision from the data itself. In masked pre-training (MPT), parts of an input are hidden and the model learns to predict them from visible context.

Masked prediction tasks

- Some observations substantially reduce predictive uncertainty.
- Others add little beyond the available context.
- Processing every candidate can therefore be redundant.

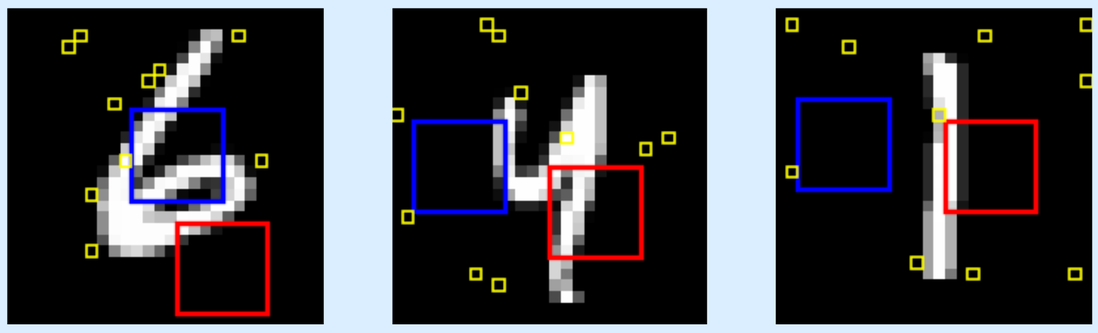
Our perspective is being able to **quantify uncertainty reduction** directly in **predictive data space**, over observable masked targets.

Masked pre-training (MPT) as a prediction task

Consider an input $x = (x_1, \ldots, x_D)$ divided into:

- **(C) : the current visible context**
- **(B) : a candidate observation batch**
- **(M) : a fixed masked target**

Does observing (B) reduce uncertainty about the fixed target (M)?



Experimental setting

- Same fixed target, different candidate observations.
- PredIG measures how much revealing x_B reduces uncertainty about the yellow target $X_{\mathcal{M}}$ given the current context x_C.
- **High PredIG ≈ informative candidate; low PredIG ≈ redundant candidate**

Predictive Information Gain (PredIG)

$$\mathrm{PredIG}_\theta(x_B \mid x_C, \mathcal{M}) = H[p_\theta(X_M \mid x_C)] - H[p_\theta(X_M \mid x_C, x_B)]$$

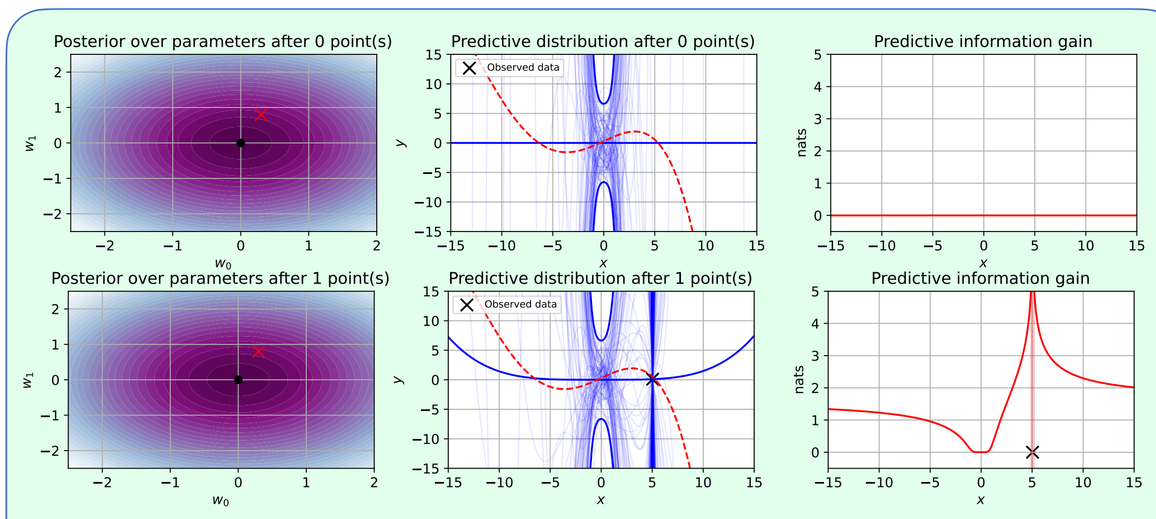
Given a complete sample (x), only the current context (x_C) is initially revealed. PredIG scores the incoming candidate (x_B) by how much it reduces predictive entropy over the same fixed target ($X_{\mathcal{M}}$). Both distributions use the same parameters (θ).

Fixed target. Fixed model. Different conditioning. Measured in **nats**.

Bayesian Linear Regression An exact sequential sandbox

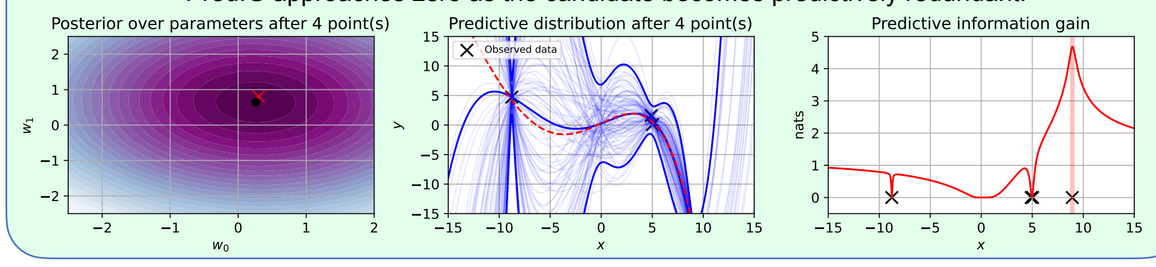
$$\mathrm{PredIG}(x) = \frac{1}{2}\log\left(\frac{|\Sigma_{*,N}|}{|\Sigma_{*,N+1}(x)|}\right), \quad \Sigma_{*,N} = \sigma_n^2 I + \Phi_* S_N \Phi_*^\top, \quad \Sigma_{*,N+1}(x) = \sigma_n^2 I + \Phi_* S_{N+1}(x) \Phi_*^\top$$

As observations accumulate, predictive uncertainty contracts and the value of additional candidates decreases.



$$\mathrm{PredIG}(x) = \underbrace{H_{\text{before}}[y_*]}_{\text{before } x} - \underbrace{H_{\text{after}}[y_*|x]}_{\text{after } x},$$

PredIG approaches zero as the candidate becomes predictively redundant.



PPCA-MPT

Masked predictive objective

$$\mathcal{L}_{\mathrm{MPT}}(\theta) = \frac{1}{|\mathcal{M}|}\log p_\theta(x_{\mathcal{M}} \mid x_{\mathcal{R}})$$

Why PPCA?

Closed-form Gaussian conditionals $p_\theta(x_M \mid x_R)$, making predictive entropy directly computable for masked prediction

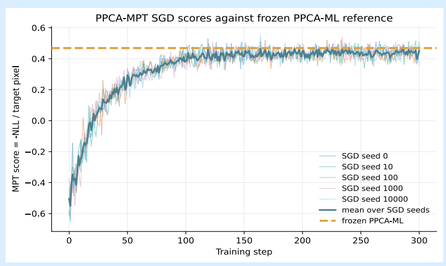
Experimental Setup

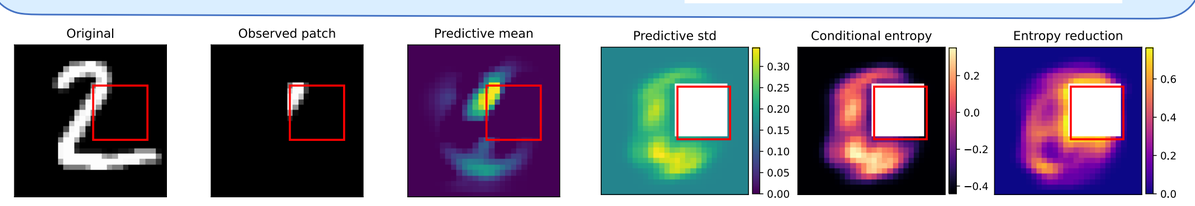
$x \in \mathbb{R}^{784}$ centered MNIST

$K = 20$, $\sigma^2 = \sigma^2_{\text{PPCA-ML}}$

[optimize only W with masked-SGD]

$|\mathcal{M}| = 200$, $|\mathcal{R}| = 584$

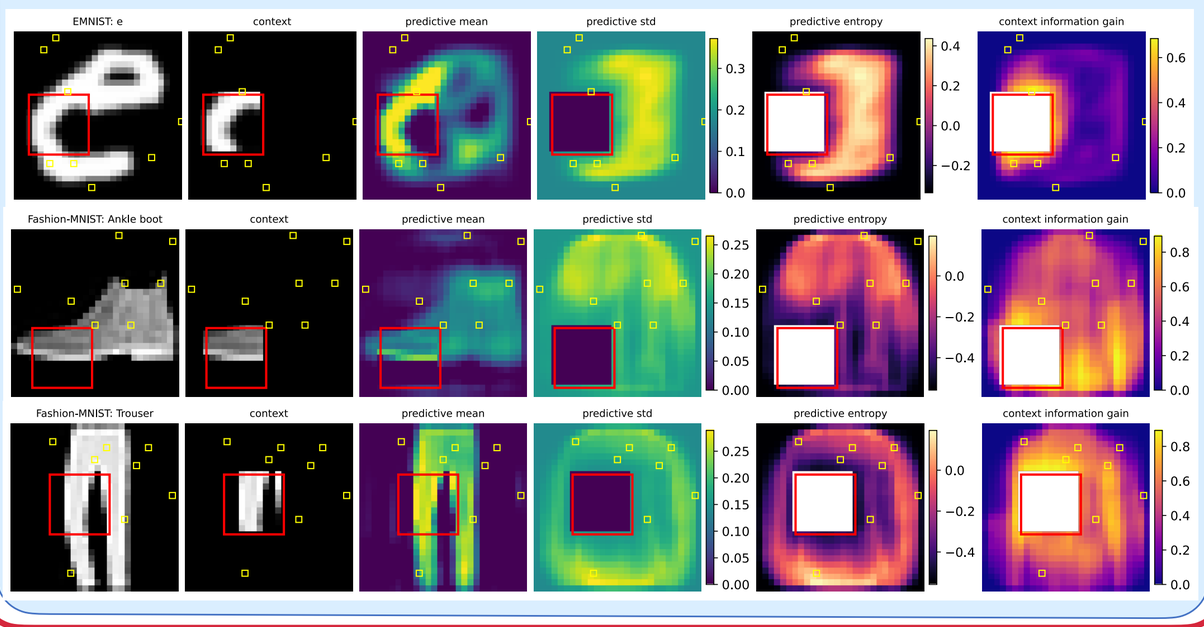




Sum of per-pixel entropy reductions over all unobserved pixels

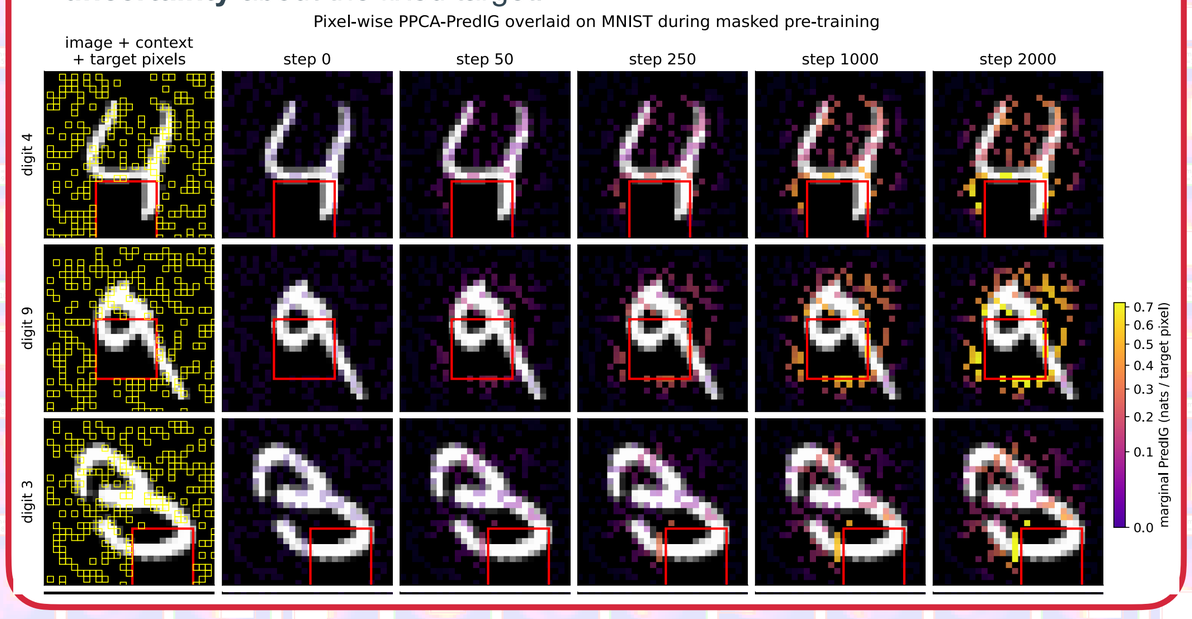
PredIG = 94.4 nats

Examples



PredIG maps during MPT

PredIG highlights which **regions** are most useful for reducing **predictive uncertainty** about the fixed target.



Discussion

- Not all masked-prediction tasks are equally informative.
- PredIG can give a principled measure of their value directly in predictive space, as predictive uncertainty reduction over a fixed target.
- This provides a principled route toward filtering redundant updates in self-supervised learning.

References

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[4] Tipping & Bishop, Probabilistic Principal Component Analysis, 1999.

[5] Advancing Deep Active Learning & Data Subset Selection: Unifying Principles with Information-Theory Intuitions, A. Kirsch, PhD Thesis, 2023

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